

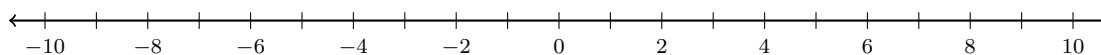
Solving and Graphing One-Step Inequalities

Adding and subtracting, multiplying and dividing, and flipping the sign when you multiply or divide by a negative

Directions. Solve each inequality, then graph the solution on the number line printed with the problem.

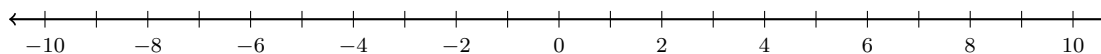
- Undo one operation, applied to *both* sides — the same move you would make for an equation.
- Multiplying or dividing both sides by a *negative* number reverses the inequality sign. Adding or subtracting never does.
- Graphing: open circle $\circ$ for $<$ or $>$, solid dot $\bullet$ for $\leq$ or $\geq$, then shade the ray toward every number that works.

1. Solve and graph: $x + 4 < 9$



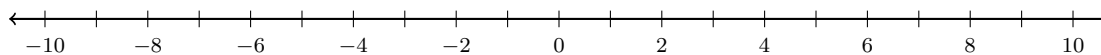
Answer: _____

2. Solve and graph: $n - 6 \geq 2$



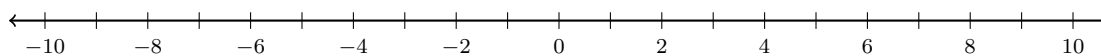
Answer: _____

3. Solve and graph: $x + 7 \leq 2$



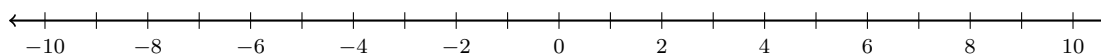
Answer: _____

4. Solve and graph: $4x > 20$



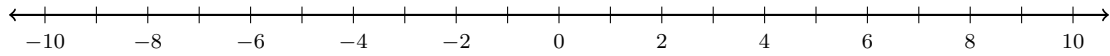
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5. Solve and graph: $\frac{m}{3} \leq 2$



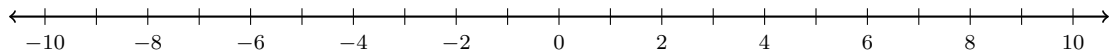
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6. Solve and graph. Watch the sign of the coefficient before you divide. $-2x > 8$



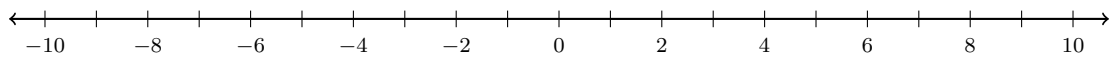
Answer: _____

7. Solve and graph: $y + 9 > 4$



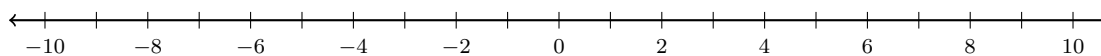
Answer: _____

8. The answer here is negative, but nothing in the solving step is. Solve and graph:
 $6w \leq -18$



Answer: _____

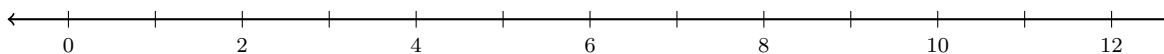
9. Solve and graph: $-\frac{x}{3} < 2$



Answer: _____

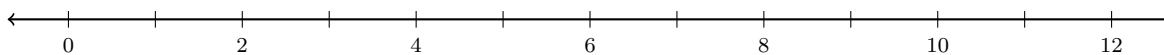
10. The chess club has 18 members and needs at least 25 to enter the district tournament. Let m be the number of new members it still signs up.

(a) Write a one-step inequality for m and solve it. Graph the solution below. (b) What is the fewest new members that will do it?



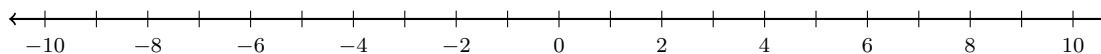
Answer: _____

11. Concert tickets cost 8 dollars each and Priya may spend at most 72 dollars. Let t be the number of tickets she buys. Write a one-step inequality for t , solve it, and graph the solution.



Answer: _____

12. Jordan solved $-5x \geq 20$ by dividing both sides by -5 and wrote the answer $x \geq -4$. Solve the inequality correctly and graph your solution. Then write one sentence saying what rule Jordan missed, and why that rule exists.



Answer: _____